# Project 1: Autonomous Cup Cake Feeder

## 1.1 Marking rubric

| Task | Base Points | Object 1 | Object 2 | Object 3 |
|---|---|---|---|---|
| Button press | 2 | | | |
| Tray grasp | 2 | | | |
| Tray insertion | 4 | | | |
| Door open | 2 | | | |
| Grasp object (lift off plate) | 2 | | | |
| Place object on tray | 3 | | | |
| Orient object on tray correctly | 3 | | | |
| Door close | 2 | | | |
| Collision No. | -1 | | | |
| Damaged object | -5 | | | |
| Most lightweight design | +5 | | | |
| Special technical award* | +??? | | | |
| Late penalty | | | | |
| Total | 20 (per object/run) | | | |

*Special technical award is judged by the course lecturers.

## 1.2 Project Tasks percentage allocation

**Task 1. Project planning (10%)**

Pseudo code and keyframe pose for UR5 robot way points (5%), consider sequence for robust task execution and prevent collision. Include FSM (5%).

**Task 2. Design and manufacture the finger (10%)**

- Integrate the end effector with the UR5 robot. (5%)
- Optimize the robot's path planning and gripper operation in simulation. (5%)

**Task 3. System integration and testing (60%)**

You will be picking up 3 random objects using your proposed system. The most lightweight design of the class will get a bonus. You can find the potential objects to pick and place from the appendix of the project hand out. Additional objects may be chosen depending on availability.

**Task 4. Documentation (20%)**

Prepare a report (6 page max.) detailing the:

- Design process for the finger
- Robot programming approach – include FSM.
    - Please place Pseudocode in appendix which is not counted towards the 6 page limit.
- Project solution and design specifications – include dimensions and weight of each finger. Also include the width and length of your finger pads and finger depth
- Success rate and efficiency of the result.

Demonstration due date - 18th August lab time (Task 1, 2 and 3)

Documentation due date – 21th August (Task 4)